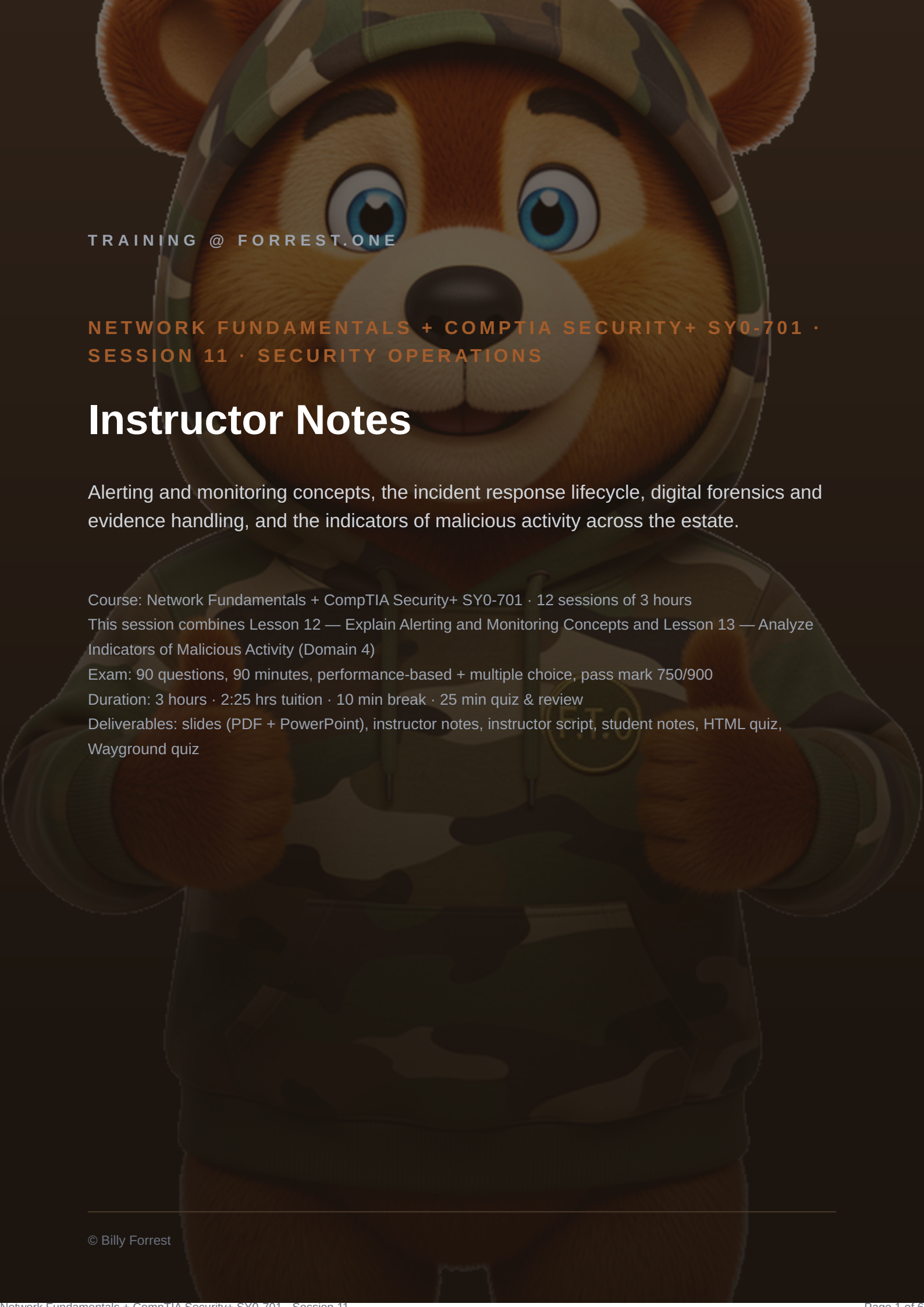




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 11 · SECURITY OPERATIONS

Instructor Notes

Alerting and monitoring concepts, the incident response lifecycle, digital forensics and evidence handling, and the indicators of malicious activity across the estate.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session combines Lesson 12 — Explain Alerting and Monitoring Concepts and Lesson 13 — Analyze Indicators of Malicious Activity (Domain 4)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Notes — Security Operations: Monitoring, Incident Response & Forensics

A 3-hour delivery guide for Session 11. Domain 4 is the largest on the exam, so this session carries weight. The session follows the expanded narration in six chapters: monitoring and network security devices, endpoint and data controls, identity and access automation, incident response, threat hunting and forensic evidence, then logs and consolidation. Keep the sequence clear: monitor, control, respond, hunt, investigate. The activities rehearse sequencing decisions, and the exam rewards order-of-operations thinking.

Session at a glance

Time	Block	Slides	Focus / activity
0:00–0:08	Welcome & objectives	1–3	Course position; six-chapter map; objectives
0:08–0:33	Ch 1 Monitoring devices	4–10	Firewall types; rules; WAF; IDS and IPS; detection methods; host logs; discussion
0:33–0:53	Ch 2 Endpoint and data controls	11–15	DNS filtering; DLP; NAC; EDR and XDR; alert workflow
0:53–1:15	Ch 3 Identity and automation	16–21	Lifecycle; SSO and federation; MFA; PAM; automation; orchestration
1:15–1:25	Break	22	10-minute break
1:25–1:45	Ch 4 Incident response	23–27	Four phases; containment; classification and roles; exercises; training
1:45–2:01	Ch 5 Hunting and forensics	28–31	Threat hunting; root cause analysis; evidence collection; chain of custody
2:01–2:13	Activity and Ch 6 Logs	32–35	Evidence activity; log sources; retention; captures, memory, metadata
2:13–2:25	Indicators	36–37	Malware table; network, application and behavioural indicators
2:25–2:50	Exam map & quiz	38–40	Indicator activity; objective map; Quiz_Session_11.html
	Close		Takeaways; checklist; Session 12 preview

Time	Block	Slides	Focus / activity
2:50–3:00		41–43	

Materials and preparation

- Projector or screen with the Session 11 slide deck (02_Slides.pdf or 02_Slides.pptx).
- One computer or tablet per pair for Quiz_Session_11.html.
- Printed or digital copy of the Student Notes for each candidate.
- Whiteboard or flipchart for the evidence-handling and indicator activities.
- An example IR plan, incident timeline, or firewall rule set (optional but valuable).
- Optional: a live firewall rule list or IDS alert log to show what real output looks like.

Delivery guidance by chapter

Slides 1–3 · Welcome (8 minutes)

Frame the session: the exam's largest domain, and the part of security that happens after prevention has already failed. Walk the six-chapter map so candidates know where the control questions and the forensic questions sit. Set the theme: evidence. What each device records, and how an analyst turns that record into a decision.

Slides 4–10 · Chapter 1, monitoring and network security devices (25 minutes)

Build the firewall progression on slide 4 deliberately: the exam tests what each generation can see. Packet filtering sees addresses, ports and protocol; stateful inspection adds the conversation; the next-generation firewall (NGFW) adds the application. Slide 5 is about process, not technology: rule creation, change control, review, and segmentation into zones including the demilitarized zone (DMZ). Slide 6 narrows to the web application firewall (WAF), which reads HTTP and HTTPS requests and blocks injection and cross-site scripting. Slide 7 draws the detection and prevention line: an intrusion detection system (IDS) informs, an intrusion prevention system (IPS) intervenes. Slide 8 compares signature-based and anomaly-based detection and lands the tuning point, since alert fatigue is a directly examined concept. Slide 9 is the analyst's lens: the xrdp log on a Kali box, and the brute-force pattern of failures then a success. Use slide 10 as the discussion, and press for the diagnosis being an alerting failure rather than a detection failure.

Slides 11–15 · Chapter 2, endpoint and data controls (20 minutes)

These controls are examined by layer, so name the layer out loud each time. Domain Name System (DNS) filtering acts on the name resolution query before the connection exists. Data loss prevention (DLP) acts on the data: in motion at the network, at rest and in use at the endpoint. Network access control (NAC) acts on the device, on a posture check before entry. Endpoint detection and response (EDR) acts on behaviour rather than signatures, which is why it catches what antivirus misses. Slide 15 gives the workflow and extended detection and response (XDR) as the correlation layer, with automated responses named: isolating an

endpoint, quarantining a file, blocking an address, revoking credentials. The closing line of the chapter is worth repeating: filtering guards the domains, DLP guards the data, NAC guards the devices.

Slides 16–21 · Chapter 3, identity, access and automation (22 minutes)

Identity is heavily examined, and it appears inside scenario questions. Slide 16 is the user lifecycle: onboarding, role change, offboarding. Draw out that late offboarding is a common audit finding and an open door. On slide 17, single sign-on (SSO) is convenience and risk at once, so pair it with multifactor authentication (MFA); federation is trust proven by a token; identity governance keeps it honest through access reviews. Slide 18 compares the three MFA factors and their weaknesses, and makes the point that two passwords are still one factor. Slide 19 is privileged access management (PAM): just-in-time access, session recording, password vaulting, and attribution. Slides 20 and 21 cover automation use cases and orchestration, then the guardrails: test changes, restrict who edits scripts, log every action, and keep a human on judgement.

Slide 22 · Break (10 minutes)

Announce what comes next: incident response, the four phases in the order the exam expects.

Slides 23–27 · Chapter 4, incident response (20 minutes)

The four phases on slide 23 are the spine of the session: preparation, detection and analysis, containment, eradication and recovery, lessons learned. Drill the order, because scenario questions lose marks on sequence. Slide 24 separates containment from eradication and names the trade-off: containment preserves evidence but can alert the adversary. Slide 25 supplies classification, severity and roles, and makes the link explicit: severity drives urgency, and named roles remove hesitation. Slides 26 and 27 cover exercises and the improvement loop. Ask which exercise type they would run first with no budget, and draw out why the tabletop is the cheapest way to find gaps in communication.

Slides 28–31 · Chapter 5, threat hunting and forensic evidence (16 minutes)

Hunting is proactive and assumes the attacker is already inside. Slide 28 gives the two methods: hypothesis-driven investigation, and indicator sweeping for addresses, hashes and domains. Slide 29 runs root cause analysis in order and stresses the final step, verifying the fix, because without it the incident returns. Slides 30 and 31 are the forensic core. Collection, write blockers, bit-for-bit imaging including deleted and hidden data, and hash verification. Then chain of custody, authorisation by warrant or consent, and privacy law (GDPR and ECPA). Say the rule out loud: analyse the copy, never the original. Add the volatility point that the indicator activity in the quiz tests: registers, then memory, then disk.

Slides 32–35 · Evidence activity and Chapter 6, logs (12 minutes)

Run the evidence activity on slide 32, then turn to logs. Slide 33 names the five sources and their tell-tale events: security logs for logins and privilege change, system logs for boot and hardware events, application logs, network device logs, and firewall logs for allowed and denied traffic. Slide 34 is collection, centralisation into a SIEM, correlation, and retention, with the consequence stated plainly: you cannot investigate what you never kept. Slide 35 adds the three sources beyond logs: network capture, memory forensics, and metadata analysis.

Slides 36–37 · Indicators (6 minutes)

The malware table is recognition level, so keep it brisk: virus and worm, Trojan, ransomware, spyware and keylogger, rootkit, botnet. Slide 37 groups network, application and behavioural indicators and lands the lesson that different monitoring layers catch different indicators.

Activities (slides 32 and 38)

Handle the evidence: RAM and open connections first (volatility); image and hash before analysis; document every handover, because a break makes it inadmissible. Read the indicators: C2 beacon, impossible travel, ransomware. Eight minutes in pairs, four minutes debrief for each.

Slides 39–43 · Exam map, quiz and close (35 minutes)

Slide 39 maps the four objectives to the chapters just taught. Run the quiz (slides 40), review the low-scoring areas, walk the takeaways on slide 41, assign the homework on slide 42, and preview the final session.

Common misconceptions and corrections

Misconception	Correction
"A firewall sees the application"	Only a next-generation firewall (NGFW) adds application awareness; packet filtering cannot
"An IDS blocks the attack"	An IDS detects and alerts; an IPS is the one that drops packets and resets connections
"Anomaly-based detection is simply better"	It catches unknown threats at the cost of false positives, so it must be tuned
"Containment is the first phase"	Preparation comes first: policy, tools, plan and trained people
"Restoring service ends the incident"	Verification, intensified monitoring and lessons learned close it
"Analyse the original drive"	Image it, hash it, then analyse the copy and preserve the original
"Everything must be captured at once"	Order of volatility decides: registers, then memory, then disk
"Logs are optional evidence"	Logs and network captures are primary forensic sources
"Retention is an IT convenience"	Retention is set by regulation and by investigative need, and short retention loses the case
"Anomaly in the log equals an incident"	Hunting and detection produce candidates; analysis and root cause confirmation follow
"DLP and NAC do the same job"	DLP controls the data wherever it moves; NAC controls the device before it enters
"Automation removes the analyst"	Automation shortens response time; humans keep judgement, exceptions and ethics

Misconception	Correction
"Two passwords satisfy multifactor authentication"	Two passwords are still a single factor: combine different factor categories
"All incidents are treated alike"	Classification and severity set the urgency, the resources and the reporting
"The analyst may start collecting freely"	Authorisation, by warrant or consent, must precede collection

Exam alignment check

- All session objectives map to Domain 4 objectives (11.1 monitoring, 11.2 response, 11.3 forensics, 11.4 indicators).
- Quiz tags map every question to its exam objective for traceable revision.
- Sequencing scenarios in the quiz mirror the exam's operational style.
- Chapter 1 to 3 content supports the monitoring and control scenarios; chapters 4 to 6 support response and forensic scenarios.